Measurement of the Thermal Conductivity of a Poor Conductor

Thermal conductivity is an important index about thermal performance of materials. A poor thermal conductor can be used for heat preservation. The objective of this experiment is to get the thermal conductivity via method of stable state.

Objectives

1. Learn to measure thermal conductivity of a poor conductor.
2. Understand the design of parameter transformation method to avoid dealing with what is difficult to measure directly.

Apparatus

Poor thermal conductor sample (rubber), copper plate×2, mainframe, balance, vernier caliper.

Theoretical principle

1. Definition of thermal conductivity

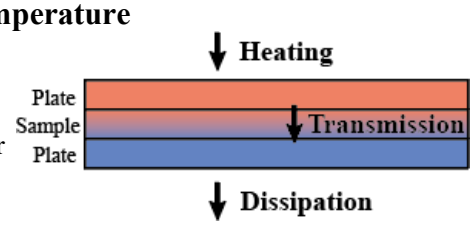
In a specific medium, if there is an ununiform temperature distribution, there is a heat transmission against the gradient of temperature. The heat goes through a unit cross-section area per unit time is proportional to the gradient of temperature. Set it as x-direction, and the cross-section area is S, then

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = -\lambda S \frac{\mathrm{d}T}{\mathrm{d}x}$$

The negative sign indicates the direction is against the gradient.

2. Measurement of cross area and gradient of temperature

Put a sample disk between two copper plates of the same area. Because copper is a good thermal conductor and its thickness h_s is much smaller than its diameter D_s, it can be approximately considered that the temperatures of the copper plates are constant separately, $T_{1\mathrm{p}}$ and $T_{2\mathrm{p}}$, and there is a gradient of temperature in the sample along the axis of the system, which is uniform on the cross-section:



$$\frac{\mathrm{d}T}{\mathrm{d}x} = \frac{T_{1\mathrm{p}} - T_{2\mathrm{p}}}{h_\mathrm{s}}$$

the temperature of the top plate $T_{1\mathrm{p}}$ is higher than the bottom $T_{2\mathrm{p}}$. The cross-section area is $S = \pi D_s^2/4$. Take care that the sample should be tightly contact with the plates, or the air between them will affect the distribution of gradient of temperature. When $T_{1\mathrm{p}}$ and $T_{2\mathrm{p}}$ stop changing, the temperature distribution of the whole system is stable, which is called stable state.

3. Measurement of the power of heat transmission

The power of heat transmission $\mathrm{d}Q/\mathrm{d}t$ cannot be measured directly. We need to transform it into what can be measured directly. At a stable state, the hotter copper plate should be heated continuously, and the colder one dissipates heat to the surroundings. The dissipating power is equal to the transmission inside the sample.

The dissipating power cannot be measured directly either, and we need to transform it again. If the colder plate is heated directly with the sample removed, its temperature will increase even further. When the temperature increases to a certain value higher than $T_{2\mathrm{p}}$, stop heating and leave it cooling in the air. Measure the change of its temperature with respect to time as a T_2-t curve in the cooling process, until T_2 decreases to a certain value lower than $T_{2\mathrm{p}}$. Then the gradient of the curve at $T_{2\mathrm{p}}$ indicates the changing rate of temperature $\mathrm{d}T/\mathrm{d}t$, and leads to the dissipating power:

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = -mC\frac{\mathrm{d}T}{\mathrm{d}t}$$

where m is the mass of the copper plate, and C is the specific heat of copper. However, take in account that the area of dissipating surface of the plate in the cooling process is not equal to that in the stable state. The former is the whole surface area, $\pi D_\mathrm{B}^2/2 + \pi D_\mathrm{B} h_\mathrm{B}$, while the latter is $\pi D_\mathrm{B}^2/4 + \pi D_\mathrm{B} h_\mathrm{B}$ because there is a face covered by the sample. Because the dissipating power is proportional to the area, the power at the stable state should be

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = -mC\frac{\mathrm{d}T}{\mathrm{d}t}\frac{\pi D_\mathrm{B}^2/4 + \pi D_\mathrm{B} h_\mathrm{B}}{\pi D_\mathrm{B}^2/2 + \pi D_\mathrm{B} h_\mathrm{B}} = -mC\frac{\mathrm{d}T}{\mathrm{d}t}\frac{D_\mathrm{B}/2 + 2h_\mathrm{B}}{D_\mathrm{B} + 2h_\mathrm{B}}$$

And the final equation of thermal conductivity derived from what can be measured directly is

$$\lambda = -mC\frac{\mathrm{d}T}{\mathrm{d}t}\frac{D_\mathrm{B}/2 + 2h_\mathrm{B}}{D_\mathrm{B} + 2h_\mathrm{B}}\frac{4}{\pi D_s^2}\frac{h_\mathrm{s}}{T_{1\mathrm{p}} - T_{2\mathrm{p}}}$$

Contents

1. Basic parameter measurement: get the diameters and thicknesses of the sample and bottom copper plate, of which take 6 measurements for each item. Record the mass of the bottom copper plate.
2. Establishing stable state: place the sample between the copper plates and start heating. As the system reaches an equilibrium, press "Enter" to stop and record the temperatures under the equilibrium: $T_{1\mathrm{p}}$=_____, $T_{2\mathrm{p}}$ =_____(℃).
3. Secondary warming: remove the sample and only the bottom copper plate is heated. Press "Quit" as the temperature of the bottom copper plate reaches $T_{2\mathrm{p}}$+10(℃).
4. Cooling: stop heating and make the bottom copper plate cooled in the air. Press "Quit" when it is shown OK on the bottom right of the monitor.
5. Collecting cooling records: get the temperatures during the cooling period. Copy a value for every three records in the table "T_2-10" (corresponding to every 30 seconds, i.e. the series numbers are 1st, 4th, 7th, …). The records should cover the range from $T_{2\mathrm{p}}$+10℃ to $T_{2\mathrm{p}}$-10℃.

Attention

1. The probe T_1 of the mainframe should be plugged in the top plate, while the probe T_2 should be in the bottom plate. Do not make it wrong.
2. Keep on heating from the establishment of thermal equilibrium to the end of secondary warming.
3. If the monitor displays nothing with a blue screen, turn the instrument off and then on again immediately.
4. You can decide whether the steps of thermal equilibrium and secondary warming are over by yourself, while the end of the cooling step must be determined by the instrument. Do not finish cooling in advance.
5. There is no need to record the temperatures in the cooling period. They can be found in the table T_2-10 after that.

Data collection

1. Basic parameters: the diameters of the sample and the bottom copper plate, D_s, D_B, the thickness of them, h_s, h_B, and the mass of the bottom plate: m_B=_____.

Table 1 The diameters and the thickness of the sample and the bottom copper plate

	1	2	3	4	5	6	Average
D_s/cm							
D_B/cm							
h_s/cm							
h_B/cm							

2. The temperatures in the cooling period:

Table 2 Temperature in cooling

Serial number	1	4	7	10	…
Time: t/s	0	30	60	90	…
Temperature: T_2/°C					…till T_{2p}-10°C

※ This table is recommended to be prepared for at least 15 minutes (30 records).

Data Analysis

1. Draw the cooling curve T_2-t, fit it by means of polynomial fitting (to the second order), and give out the fitting equation. Find the cooling rate $\mathrm{d}T/\mathrm{d}t$ at T_{2p} by the equation.
2. Get the thermal conductivity of the sample:

$$\lambda = -mC\frac{\mathrm{d}T}{\mathrm{d}t}\frac{D_\mathrm{B}/2 + 2h_\mathrm{B}}{D_\mathrm{B} + 2h_\mathrm{B}}\frac{4}{\pi D_s^2}\frac{h_\mathrm{s}}{T_{1\mathrm{p}} - T_{2\mathrm{p}}}$$

   where C=385J/(K·kg).
3. Calculate the theoretical value, and determine the error by comparing the experiment result with that:

$$\hat{\lambda} = \lambda_0\left(\frac{T_{2\mathrm{p}}}{273.15\mathrm{K}}\right)^{3/2}$$

   where λ_0=0.140W/(m·K), T_{2p} is in K.

Take note in data analysis:

1. The transformation between Celsius temperature and absolute temperature is T(K) = T(°C)+273.15
2. Take seven quantities in consider when calculating the uncertainty: m_B, D_s, D_B, h_s, h_B, T_{1p}, T_{2p}. Because $\mathrm{d}T/\mathrm{d}t$ is from a nontypical fitting, you can neglect that.
3. Take care of the indicating error of instruments involved: 0.1g for the balance, 0.02mm for the vernier caliper, and the temperature on the mainframe: 0.25°C (equivalent to 0.25K).

Question

The conductivity given by the experiment is usually smaller than the real value. What system errors cause that?